AD PROJECT INSTRUCTIONS

AD Project Manager: Dr Vincent Chung

Chief GDipSA: Chia Yuen Kwan

Important Note

This document contains a lot of information and guidance. Please make sure that you read it carefully before your main submission at the end of Sprint 3, especially those with important and/or highlighted words.

Objectives of AD Project

- Provide an opportunity for practising the technical and management skills acquired throughout graduate diploma programme.
- Prepare students for internship.
- Equip and nurture students with skills to:
  - Identify real world problems and how to resolve them using technologies.
  - Explore, propose and define scope of work in the form of digital products.
  - Exercise their creativity and propose innovative ideas rather than fixed scope and content.

Lecturers

- Dr Esther Tan
- Mr Chia Yuen Kwan
- Dr Liu Fan
- Mr Chandra
- Ms Chen JunHua
- Mr Kenneth Phang

- AD Project Manager and Overall Coordinator
  - Dr. Vincent Chung

Overall Timeline and Deliverables

Prepare Project Proposal

Present Project Proposal

Refine the requirements

Demonstration of the final product

Inception & Requirements Sprint:
1) Defining problem and proposal.
2) Present the proposal.
3) Refine the requirement model

1st Sprint – Design & Prototyping
Design & Prototyping by 2nd week

2nd Sprint
MVP by 3rd week

Final Sprint
MVP, code walkthrough, by 4th week

Refine & Touch up deliverable over the enrichment week

A **recorded video presentation** of the team deliverables

Project Activities

Activities	Component	Stakeholders
Sprint 0: week 1 (Inception & Requirements)	1) Define problem statements, present proposals with high level requirements Problem statement (scenario), solution proposal, scope & technologies, high-level requirements	AD Project Manager and coordinator
	2) Refine & prioritized requirements	Team Advisor
Sprint 1: week 2 (Design & Prototyping)	Design finalisation, prototype, and setting up of infrastructure	Team Advisor
Sprint 2: Week 3 (Development)	Working Features, UAT, DevSecOps (CI/CD pipeline)	Team Advisor
Sprint 3: Week 4 (Development)	Working Features, UAT, code-walkthrough, DevSecOps, product demonstration video	Team Advisor
Final Presentation & Demonstration		Assessors
Showcase	Packaging showcase materials	AD Project Manager

Project Proposal to be provided by students

- Project Title <u>(Please ensure that the proposed project is original. Projects that have been implemented in previous CA Projects will not be accepted.)</u>
- Team members, matric numbers and emails
- Business Requirements
  - A reference to the problem you are trying to resolve
  - A high-level use case diagram
- Technical Requirements
  - The proposed solution should incorporate the following technologies as part of an integrated solution:

Agentic AI	Mobile Application	Web Application	Machine Learning	Data Visualisation (UI charts/dashboard)

    - All components should be integrated through a common backend to demonstrate a cohesive end-to-end solution
  - For <u>each use case</u>, specify the technologies involved. For example:
    - Mobile Client - Android Native / React Native / Java Spring Boot…
    - Web Client – Angular/React.js/ HTML / CSS/ Java Script
    - Backend Server side - .Net MVC Core / Java Spring Boot ….
    - DB – My SQL Server, MS SQL Server …
    - Machine Learning – Indicate the machine learning models
    - UI Storyboarding - Chart.JS / JavaScript Charting Libraries ….
  - One or more features shall be deployed on the cloud

Sprint 0 – Product Vision

- **Objective**
  - You will work in teams to design a **digital solution** based on a **common theme** provided by the instructor. The goal is to simulate a **hackathon-style problem-solving experience**, focusing on **ideation, requirements analysis, and solution design**, rather than full system implementation.

Sprint 0 – Product Vision

- **Problem Theme**
  - A **common theme** will be provided by the instructor.
  - Your team is expected to:
    - Interpret the theme
    - Identify a meaningful problem within the theme
    - Propose a digital solution that addresses the problem

Sprint 0 – Product Vision Showcase

- **Present the followings to showcase your digital solution**
  - 1) Solution Overview (High-level)
    - Problem statement
    - Target users
    - Key objectives of the solution
    - How the solution address the given theme

Sprint 0 – Product Vision Showcase

- 2) High-level Use Case Diagram
- 3) Prioritized Product Backlog
- 4) Prototype Screens (UI Mockups)
  - How users navigate the prototype scrrens
  - Tools:
    - You may use AI-assisted tools such as Figma or other AI design tools (for Sprint 0 Prototype screens showcase **<u>only</u>**)
    - Note: You are ***<u>not</u>*** developing the full-scale application in this sprint!

Deliverables – End of Sprint 0

Work with advisor

Deliverable and artifact include the followings:

- High level use case diagram
- Prioritised Product Backlog
- User Interface Design
- **Individual Contribution Report**
- **Project Status Report**

Project team may produce additional artifacts relevant to the project

Deliverables – Sprint 1

Work with advisor

Typical deliverable and artifact include the followings:

- Sprint Backlog
- Overall Software Architecture, includes
  - Database Design
  - Logical Data Model
- Software Prototype
- **Individual Contribution Report**
- **Project Status Report**

Project team may produce additional artifacts relevant to the project

Prototyping Strategies

- Purpose
  - To confirm the team has the competence to deliver the features using the technologies chosen
  - Prototype – A template for actual development
- Types of prototype
  - Option 1. Throwaway prototype
    - The prototype will not be reused for next sprint
  - Option 2. Evolutionary prototype
    - Expand the prototype as actual development in subsequent sprints
- What features should I select for prototyping?
  - A simple feature from the different group of targeted technologies

Deliverables – Sprint 2

Work with advisor

Typical deliverable and artifact include the followings:

- Sprint Backlog
- Design *relevant* for the selected features
    - E.g. sequence diagram, class diagram
- Working features
- Individual Contribution Report
- Project Status Report
- DevSecOps (CI/CD pipeline)

Project team may produce additional artifacts relevant to the project

Deliverables – Sprint 3 – Main Submission

Important

Main submission should include the followings:

1. Recording of product demonstrations
2. Source code
3. Slides
   - Must include Architectural Diagrams and ER Diagrams
   - DevSecOps architecture (CI/CD Pipeline), Security test report
4. Sprint Backlog
5. Project Status Report
6. Individual Contribution Report
7. Screenshots of working features
8. Any design relevant for the selected features
   - E.g., sequence diagrams…

Project team may produce additional artifacts relevant to the project

Video Recording

Important

- Final deliverables of the refined software product
- **Video recordings of project demonstrations**
  - Each team need to submit a screen recording of their project demonstrations.
  - Teams can choose **all or only some** member(s) to present.
  - The video should **<u>not take more than 30 minutes</u>**.
  - The demonstrations should **highlight features** and **show they are working**.
  - The video should be in MP4 format.

Demo Presentation Tips

Important

- Key: ask yourself – you are **selling** your product; how can you make people **buy** it?
- Create demos for **different stakeholders**: end-users and technical reviewers.
- **Explain** how your ML models work.
- **Distinguish** between **Reports** and **Dashboards.**
- **Present** the **final product-as-a-whole**, not a collection of non-related components.
- **Highlight** at least 2 or 3 **interesting** and/or **challenging technologies/technical aspects** implemented in your product.
- Add a **summary page** about main **product features** and **implemented technologies** at the start/end of your presentation.

Source code

Important

- Submit your source codes for all platform into a folder.
- Include any necessary database files for the product.
- Include a short ***readme.txt file*** how to run the product **locally**.

How to submit?

- Zip everything into a file named ***Teamxx AD Deliverables.zip.***
    - For example: *Team01 AD Deliverables.zip*
- Submit the file into the respective folder in Canvas.
- If the submission is too large, please resize your video and/or zip your files accordingly.
- **One team** only needs to have **one submission** by a representative member.

Some important notes

Important

- For videos and presentations
  - Do **prepare** your **scripts.**
  - Do **rehearse** a few times before the final presentation and submission.
  - **Review** your video before submitting.
- For source code
  - **Review** your source code before submission.
- Review your final submission
  - Consider asking some other teammates to check.

Individual Contribution

- Each student will be assessed based on both **teamwork** and **individual** efforts.
- The students' performance will be **measured** throughout the **entire project duration** and **during** all **contact hours** with the **lecturers**.
- 100% of daily contribution to the during AD project is expected.
    - **Important: 100% attendance in Singapore is required.**

Students shown to have very low contribution to the project in any way will be given a failing grade regardless of the marking scheme.

Peer Evaluations

- Each student will evaluate his/her team-mates via an online Peer Evaluation Form.
- The purpose of the evaluation is to **present each student's perception** of **how** his/her **team-mates have performed** during the project.
- In general, students will be evaluated on their **leadership**, **technical competency**, and **teamwork**.
- **Important:**
  - It's **student responsibilities** to submit the Peer Evaluation Form on time.
  - **Students will be penalized if failing to submit it on time.**

Students shown to have very low contribution to the project in any way will be given a failing grade regardless of the marking scheme.

Assessment Matrix

Section	Assessments	Deliverables	Marks	By	Timeline
A	Feasibilities Study and Product Usability	*Product Features*	20	SA Lecturers	Product Vision Showcase and Final Product Presentation
B	Project Artifact	*Software design components, sprint planning, backlog and review*	15	Team Advisor	End of the Project
C	Software Quality	*Quality of Code*	20	SA Lecturers	End of the Project
D	Product Features	*Product Features*	25	Team Advisor	Sprint 0,1, 2, 3
E	DevOps	*CI/CD pipeline, Security testing*	10	SA Lecturers	Sprint 2, 3
F	Individual contribution and performance	*Peer Evaluation*	10	Peers	Before end of project
		Total	**100**		
	Individual Professionalism	*Daily contribution report and conduct*	Team marks will be adjusted based on individual contributions and performance. **Team advisors may give individual marks**		

Project Advisory Support

- Each team will be assigned with a SA lecturer as an advisor.
- Arrange with your advisor for any meeting or consultation needed.
- **Important: be proactive!** Send an email to introduce yourself and arrange meetings with your advisor regularly (at least once per week)

Schedule

Schedule	Where to upload artifact	When
Proposal Presentation	MS Teams	Middle of Week 1
Sprint 0	MS Teams	End of Week 1
Sprint 1	MS Teams	End of Week 2
Sprint 2	MS Teams	End of Week 3
Sprint 3	MS Teams	End of Week 4
Main Submission	Canvas	End of Week 4
Final Presentations & Demonstrations	Face-To-Face	End of Week 4
Post Submission - Product Showcase	Canvas	Within 1 week after AD Project completion

Proposal Presentation

- Present your proposed digital solution
- Brief description of your problem statement
- High level use case diagram
- Technology implementation for each use cases

Final Presentations & Demonstrations

Important

- Each team will have several **presentations**
- **Every member must attend**
- **Every** member must be ready to present your own part
- Team members should be able to explain their code implementation

Because time is limited, please come prepared with everything properly set up and there will be no time given for testing/trial run, re-presentation, etc.

Presentation & Demonstration

Important

Time/Venue: TBC

Product Inspection	Software Quality	DevSecOps
Review user friendliness of the product, flows of features, including: • Overall functionality and seamless flow of use cases • Ease of use, fulfilling product's business objective, completeness of solution	Review the technical aspects of the product, including: • Overall functionality and seamless flow of use cases • Implementation and integration of technologies • Implementation best practice • Exception handling & validation	Review the implementation of the DevSecOps practices: • Secure coding practices • CICD pipeline • Automated testing tools • Security testing – Remediation of the issues
Preparation • Product demonstration • All deliverables and artifacts (e.g., Product Status report)	Preparation • Product demonstration • Highlights of technical implementations	Preparation • CICD pipeline Demo • Automated test reports • Security test report, Remediation, final report

Post Submission - Product Showcase

- This submission is **separate** from the main submission at the end of Sprint 3
- Objective: help you to reflect and create a refined version of your product to **improve your employability**
- To be done during <u>enrichment week</u>
  - Usually ~ 1 week after your Sprint 3 submission
- Deliverables
  - **10 mins videos** (could be taken from the same video submitted at the end of sprint 3 if there are no updates)
    - Emphasize on the advanced and prized feature such as dashboard, recommender etc.

Note: marking is based on Sprint 3 Submission, **not** this Product Showcase. However, your mark will be **<u>deducted</u>** without this submission.

Post Submission - Product Showcase

- Deliverables
  - **Consolidated Group Report**
    - Product Introduction

      (The system developed might not be the exactly the same as what was proposed)
    - Background
    - System features developed
      - Include high level use case diagram or product backlog Consolidated Group Report
    - Recommendations
      - Any recommendations for future phases?
    - Lesson Learned

Post Submission - Product Showcase

- Deliverables
  - **Individual Reflection (Individual Submission)**
    - Roles and Responsibilities
    - Major Challenges & Lesson Learned
      - Technical aspects
        - 1 or 2 Coding Bug(s)/Problem(s) Encountered
          - Codes with issues relating to either (1) performance, (2) reliability, (3) good coding / CICD habits, or (4) elegant coding
          - Including in your reflection
            - The codes (selected, not copy-and-paste everything)
            - Description of the situations
            - How you resolved/overcome/tackled the issues
      - Non-technical aspects
        - Could be related to development strategies, communication with the team
        - Describe the problem and any resolution

AD Project Tips

- You may want to do great things, but beware the time limit
  - However, the project should not be too trivial, either
- After week 2, make sure you always have an MVP
- **At least**, your final product should include
  - A Web portion
  - A Mobile portion
  - A Python ML portion
- The final product should be a **work-as-a-whole**, not a collection of **non-related components**
- Work in teams, do not solo
- If you choose ReactJS or NodeJS, make sure you have done enough research

Proritization Strategies

- Remember the Goals
  - Complete business process
  - Demonstrate **implementation** and **integration** of different technologies
    - Basics
      - Mobile / Web
      - Python ML
      - Cloud Deployment
    - Advanced: make use of 3rd party API or technologies not learnt...
- Recommended prioritization strategies
  - **Business values** to the workflow
  - **Variety of technologies: depth and breadth**

Admin Matters

- Canvas
  - Ensure that you can view AD Project module in Canvas
  - Mainly used for submission
- MS Teams
  - Ensure you are being assigned the correct Teams
  - Mainly used for team communication
- If you face any admin issues, email the coordinator and CC all your members
- For any special applications/requests, email the coordinator and CC the program chief

Locations

- You don't need to be physically in class all days
- You are **required to be with the whole team** when meeting the advisors (4-6 meetings)
  - The respective advisor will decide if it's F2F or through Zoom
  - The advisor is encouraged to meet the team F2F at least one time, preferably the first meeting
- The school provides a smaller classroom for teams who want to work on their AD Projects at school
  - The classroom is not the same for all days, please check the timetable or information screen at the main lifts
  - Teams are **strongly encouraged to sit together often** to improve productivity

Appendix – Secure Coding Practices

- You should demonstrate that your applications are secure under various security threats. For example:
  - Broken Authentication
  - Improper Session Termination
  - Sensitive Data Exposure
  - Injection
- References:
  - https://cheatsheetseries.owasp.org/cheatsheets/DotNet_Security_Cheat_Sheet.html
  - https://owasp.org/www-pdf-archive/OWASP_Code_Review_Guide_v2.pdf

Questions?